KENYA MEDICA TRAINING COLLEGE

FACULTY OF REHABILITATIVE SCIENCES

PORT REITZ CAMPUS

DEPARTMENT OF ORTHOPAEDIC TECHNOLOGY

CONTINUOUS ASSESSMENT TEST

FOR

DIPLOMA IN ORTHOPAEDIC TECHNOLOGY

Paper: Material Science **Class:** September 2023

Date: 20/05/2024 **Time:** 8:30am-9:30am

INSTRUCTIONS

Attempt all questions

1. Describe the chemical reaction that occurs when plaster of Paris sets. What properties does this setting reaction impart to the material? (5 marks)
2. What are some factors to consider when selecting a metal for a specific application? (5 marks)
3. In your opinion, how can a basic understanding of material science be beneficial in everyday life and in the field of orthopaedic technology? (5 marks)
4. Describe the common processes used in heat treatment processes of metals. How do these methods affect the material's properties? (10 marks)
5. Why are some materials good insulators, while others are conductors? (5 marks)
6. Discuss atleast five (5) factors that affect the setting time of plaster of Paris (10 marks)

ACE Scanner